

1. Administrative & Study Identification

Full Study Title: Safety, Tolerability and Pharmacokinetics of AZD1775 (Adavosertib) Plus MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumours **Short Title/Acronym:** Adavosertib + Durvalumab in Advanced Solid Tumours **Protocol Number:** D6015C00002 **NCT Number:** NCT02617277 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD1775 (Adavosertib) and MEDI4736 (Durvalumab) **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, tolerability, and pharmacokinetics of oral therapy with AZD1775 (adavosertib) when combined with a fixed-dose of MEDI4736 (durvalumab), and to determine the maximum tolerated dose (MTD) and recommended Phase II dose (RP2D). **Secondary Objectives:** To determine the immunogenicity, pharmacodynamics, and preliminary anti-tumour activity (including ORR, PFS, and OS) of the combination in patients with refractory solid tumours.

2.2 Background & Rationale To address the gap in treatment for patients with advanced, refractory solid tumours. WEE1 inhibition (adavosertib) abrogates cell cycle arrest and may sensitize tumours to immune checkpoint blockade. This study explores a novel combination of a WEE1 kinase inhibitor and a PD-L1 inhibitor (durvalumab) to identify a safe dosing schedule and evaluate potential synergistic anti-tumour activity.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Dose-escalation) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** Approximately 56 evaluable patients **Number of Sites:** Multicenter **Estimated Study Start:** November 2015 **Estimated Primary Completion:** Q4 2018

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males and females ≥ 18 years of age; Weight ≥ 30 kg. 2. Histologic confirmation of a solid tumour (excluding lymphoma) refractory to standard therapy or for

which no standard of care regimen exists. 3. Measurable or non-measureable disease according to RECIST v1.1, with an ECOG Performance Status of 0-1. 4. Adequate baseline organ function (e.g., ANC $\geq 1500\$/\mu\text{L}$, HgB $\geq 9\text{ g/dL}$, Platelets $\geq 100,000\$/\mu\text{L}$, ALT/AST $\leq 2.5\times\text{ULN}$).

4.2 Exclusion Criteria 1. Use of approved treatment (e.g., chemotherapy, targeted therapy, biologic therapy) $\leq 21\text{ days}$ prior to the first dose of study drug. 2. Current or prior use of a WEE-1 inhibitor or any immunosuppressive medication (e.g., anti-PD-L1, anti-PD-1) $\leq 14\text{ days}$ prior to the first dose of MEDI4736. 3. Major surgical procedures $\leq 28\text{ days}$ prior to beginning study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple dosing schedules (A, B, C, D) are evaluated. Durvalumab is administered at a fixed dose of 1500 mg on Day 1 of each 28-day cycle. Adavosertib is administered BID on intermittent days (e.g., Schedule B: Days 15-17 and 22-24; Schedule C: Days 8-10, 15-17, 22-24). **Route of Administration:** Durvalumab: Intravenous (IV); Adavosertib: Oral (PO). **Duration of Treatment:** 28-day cycles, continuing until objective disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Dexamethasone administered as an anti-emetic on the first day of adavosertib consecutive dosing blocks. **Prohibited:** Other anti-cancer therapies (chemotherapy, immunotherapy, biological therapy) and systemic immunosuppressive medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Labs, ECOG, Tumor Imaging
Lead-in Phase	Days -7 to -1	(For Schedules B & C) 7-day AZD1775 lead-in for serial PK measurements
Treatment Cycles	28-Day Cycles	Durvalumab infusion (Day 1), Adavosertib dosing (specific days), Safety monitoring, PK sampling
End of Study	Follow-up	Disease progression assessment, Survival monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile, assessed by the incidence of Dose-Limiting Toxicities (DLTs) and determination of the Maximum Tolerated Dose (MTD) / RP2D. **Measurement Method:** Clinical and laboratory evaluation graded per

NCI CTCAE v4.0 (e.g., DLTs such as Grade 4 hematologic toxicity for ≥ 7 days or non-hematologic toxicity $\geq$ Grade 3).

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) and Progression-Free Survival (PFS) evaluated per RECIST v1.1. * Overall Survival (OS). * Pharmacokinetics (e.g., Cmax, AUC) of adavosertib and durvalumab.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of AEs through clinical and laboratory assessments, specifically monitoring for gastrointestinal (nausea/diarrhea) and hematological toxicities. **Serious Adverse Events (SAEs):** DLTs are strictly defined (e.g., AST/ALT $\geq 5\times$ but $\leq 8\times$ ULN not resolving to Grade 2 within 5 days) and must be reported immediately. **Data Monitoring Committee (DMC):** Utilizes a standard dose-escalation review committee to decide on cohort progression and dose modifications.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set encompasses all patients receiving at least one dose of the investigational product. **Sample Size Justification:** Utilizes a standard Phase I dose-escalation design (e.g., 3+3 design) requiring approximately 56 patients across various schedules and cohorts. **Handling of Missing Data:** Patients dropping out before completing the DLT observation period for reasons other than toxicity are replaced. Survival endpoints are censored at the last recorded date the subject was known to be alive.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical site monitoring by AstraZeneca to ensure protocol adherence, correct RECIST application, and PK sampling timing. **Audit Trail:** eCRF locking and comprehensive data clarification mechanisms to track dose adjustments and DLT occurrences.

11. Study Identifiers & Governance

Primary ID: D6015C00002 **Registry Number:** NCT02617277 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Safety, Tolerability and Pharmacokinetics of AZD1775 (Adavosertib) Plus MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumours **Official**

Regulatory Title: Safety, Tolerability and Pharmacokinetics of AZD1775 (Adavosertib) Plus MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumours

12. Clinical Framework (Advanced Solid Tumors Specific)

Primary Condition: Advanced Solid Tumors **Diagnosis Threshold:** Histologically confirmed refractory solid tumor lacking standard of care options **Medical Methodology:** Immune checkpoint inhibition (anti-PD-L1) combined with DNA damage response/cell cycle regulation inhibition (WEE1 inhibitor) **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Sequential cohort dose-escalation evaluating multiple dosing schedules (Schedules A, B, C, D) **Education Protocol (HCP):** Site training on strict DLT criteria, required anti-emetic administration (dexamethasone), and PK sampling timelines **Education Protocol (Patient):** Strict diary adherence for at-home oral adavosertib dosing and recognition of gastrointestinal or hematologic AEs **Quality Audit Mechanism:** Continuous safety reviews prior to dose escalation

14. Observation & Follow-Up Schedule

Total Duration: Continued until objective disease progression, unacceptable toxicity, or withdrawal **Onsite Visit Cadence:** Frequent visits during Cycle 1 (including lead-in days for PK), followed by Day 1 visits for 28-day cycles **Remote Monitoring Frequency:** Routine survival follow-up after treatment discontinuation **Checklist Review Interval:** Disease assessment via imaging (RECIST v1.1) performed at predefined intervals throughout the 28-day cycles

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				